

Ethanol Blending in Gasoline – Hungary

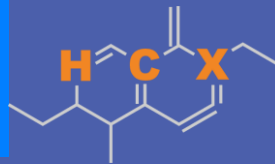
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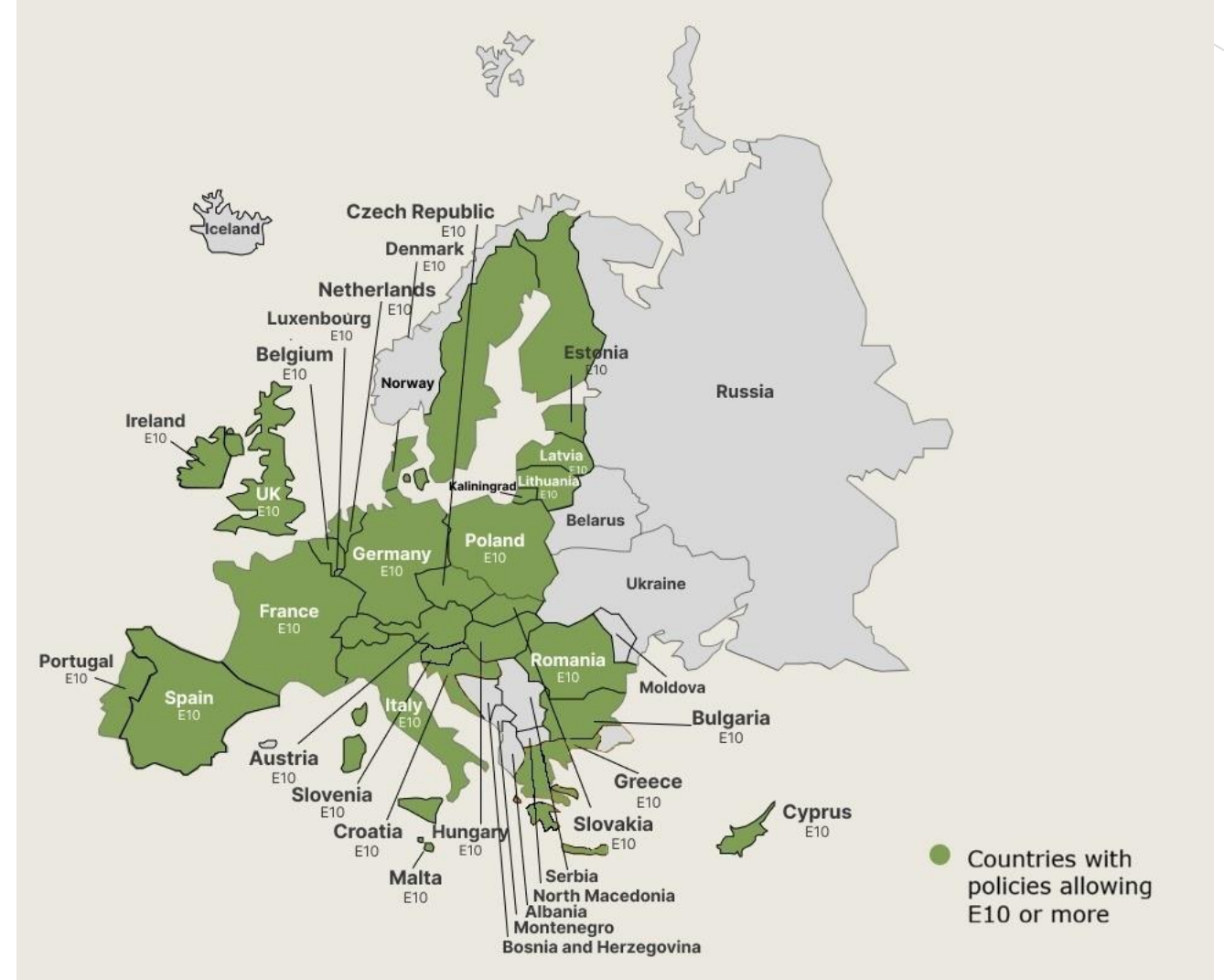


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Ethanol Blending in Europe

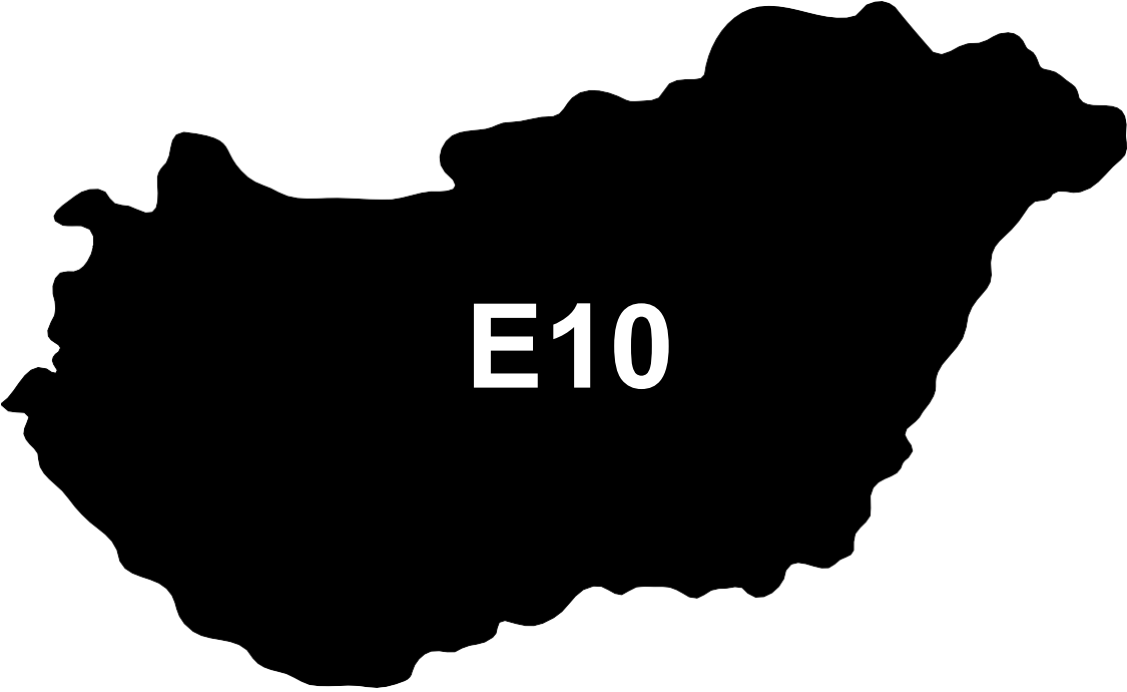
There are important fuel quality and environmental impact of vehicle emission challenges in the Region.

- The use of ethanol improves gasoline quality and creates flexibility in gasoline production.
- Ethanol use is a cost-effective way to increase gasoline octane and to replace more expensive gasoline components.
- Ethanol contributes to transport decarbonization and air quality improvement.
- There are opportunities across Europe to increase the ethanol blend level and implement new policies on the use of gasoline-ethanol blends.



Twenty eight countries with potential and additional use of ethanol were studied: 1) Gasoline market profiles; 2) Optimization of gasoline blends with ethanol and, 3) Environmental impact of gasolines blended with ethanol.

Ethanol Gasoline Blending in Hungary



E10

In 2024, gasoline consumption in Hungary was 1,544 million liters (408 million gallons) and growth rate was 0.8%. Gasoline production and net imports were 1,701 and 79 million liters (449 and 21 million gallons) correspondingly. Hungary complies with the European gasoline standards and offers two main grades: RON 95 and RON 98 with an average octane of 95.9 RON.

In 2024, ethanol consumption in Hungary was 147 million liters (39 million gallons) representing 9.5% of gasoline demand. Ethanol production and Net Imports were 609 and -462 million liters (161 and -122 million gallons) correspondingly. Ethanol source is mainly Corn based.

Source: Eurostat, European Commission

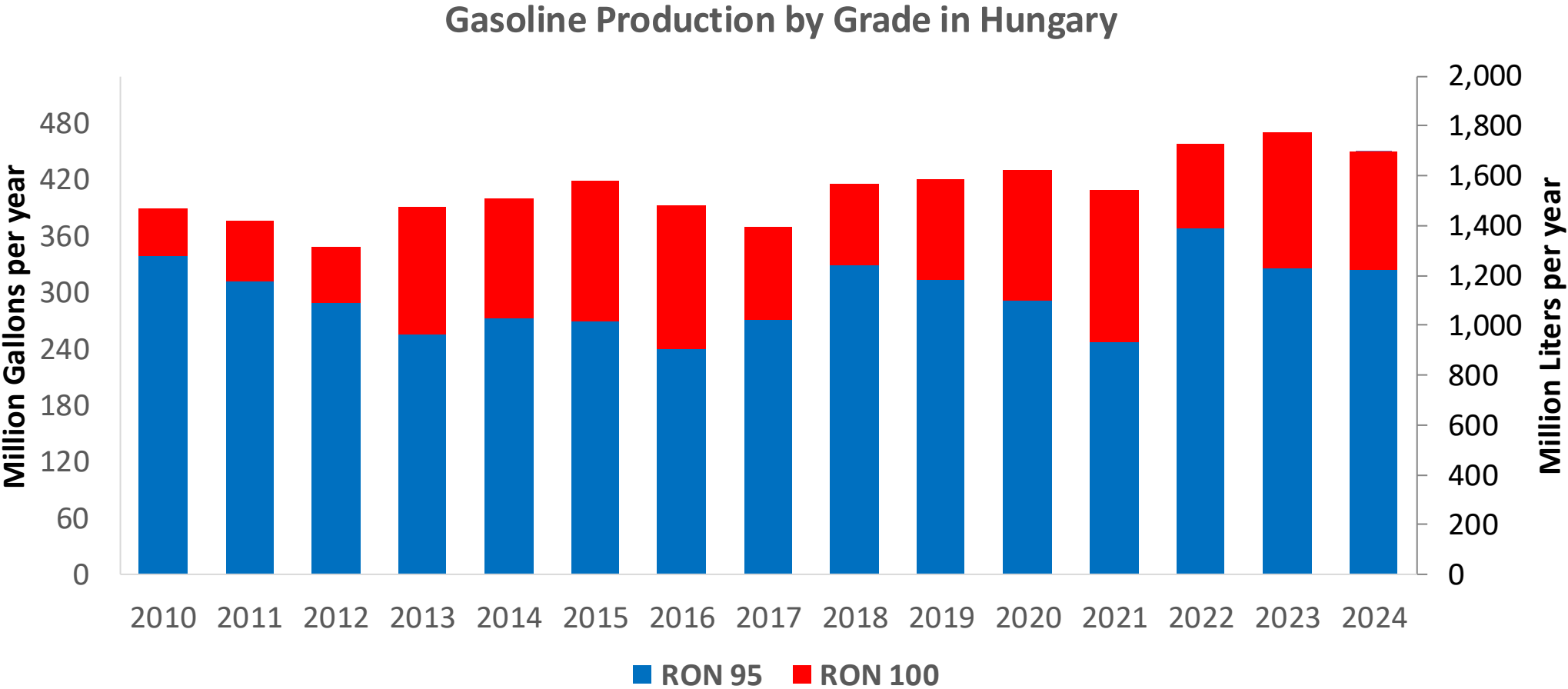
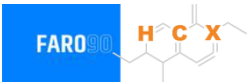
The FARO90 logo is shown in a blue box, followed by a chemical structure of a substituted cyclohexadiene. The structure features a six-membered ring with two double bonds. Substituents include a hydrogen atom (H), a methyl group (C), and an ethyl group (X) on one carbon, and a methyl group on an adjacent carbon. The FARO90 logo is positioned to the left of the structure.

The chart displays the annual consumption of RON 95 and RON 100 gasoline in the United States from 2010 to 2024. The left Y-axis measures consumption in Million Gallons per year (0 to 480), and the right Y-axis measures consumption in Million Liters per year (0 to 2,000). RON 95 is represented by blue bars, and RON 100 is represented by red bars. The total consumption is the sum of the two.

Year	RON 95 (Million Gallons per year)	RON 100 (Million Gallons per year)	Total (Million Gallons per year)
2010	355	10	365
2011	345	10	355
2012	315	10	325
2013	295	10	305
2014	325	10	335
2015	325	20	345
2016	325	10	335
2017	335	10	345
2018	350	10	360
2019	365	40	405
2020	295	85	380
2021	300	75	375
2022	415	55	470
2023	345	65	410
2024	345	60	405

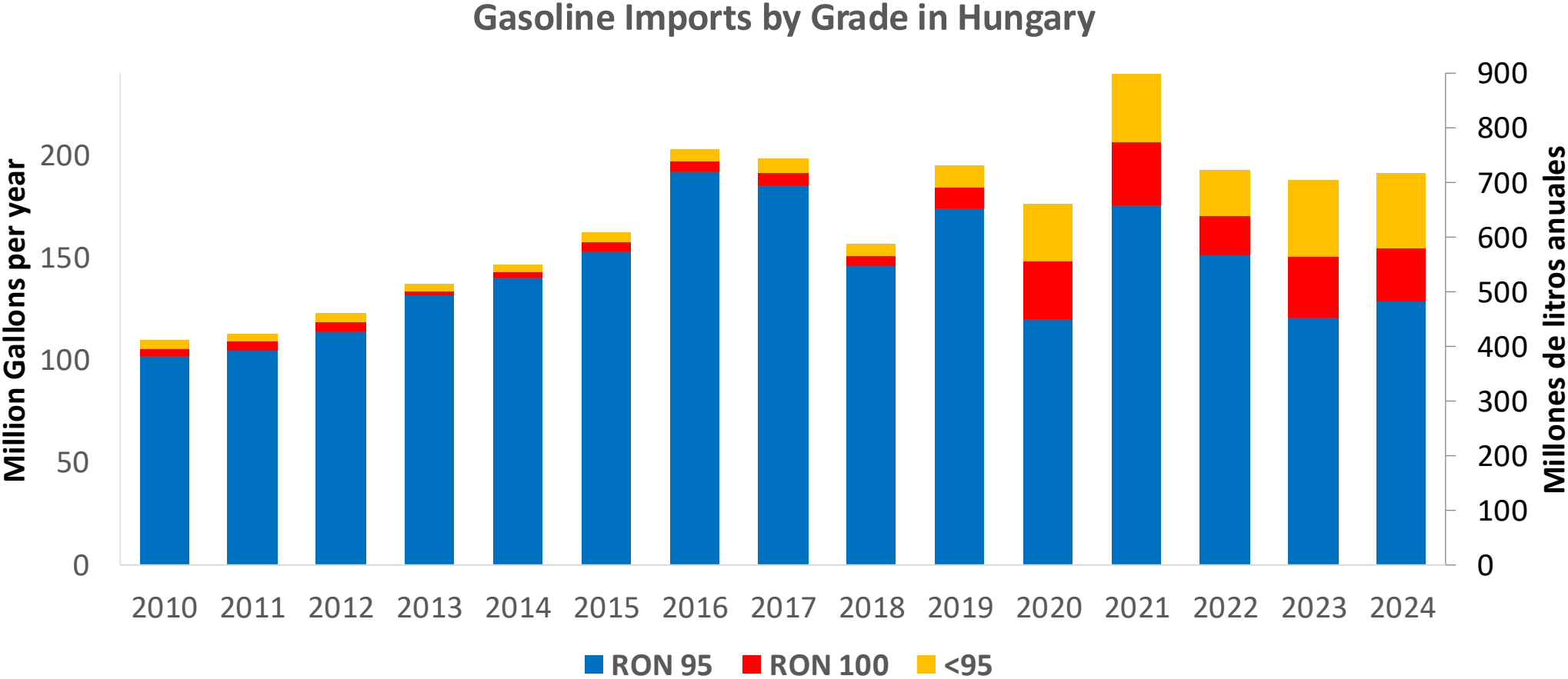
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Gasoline Production in Hungary



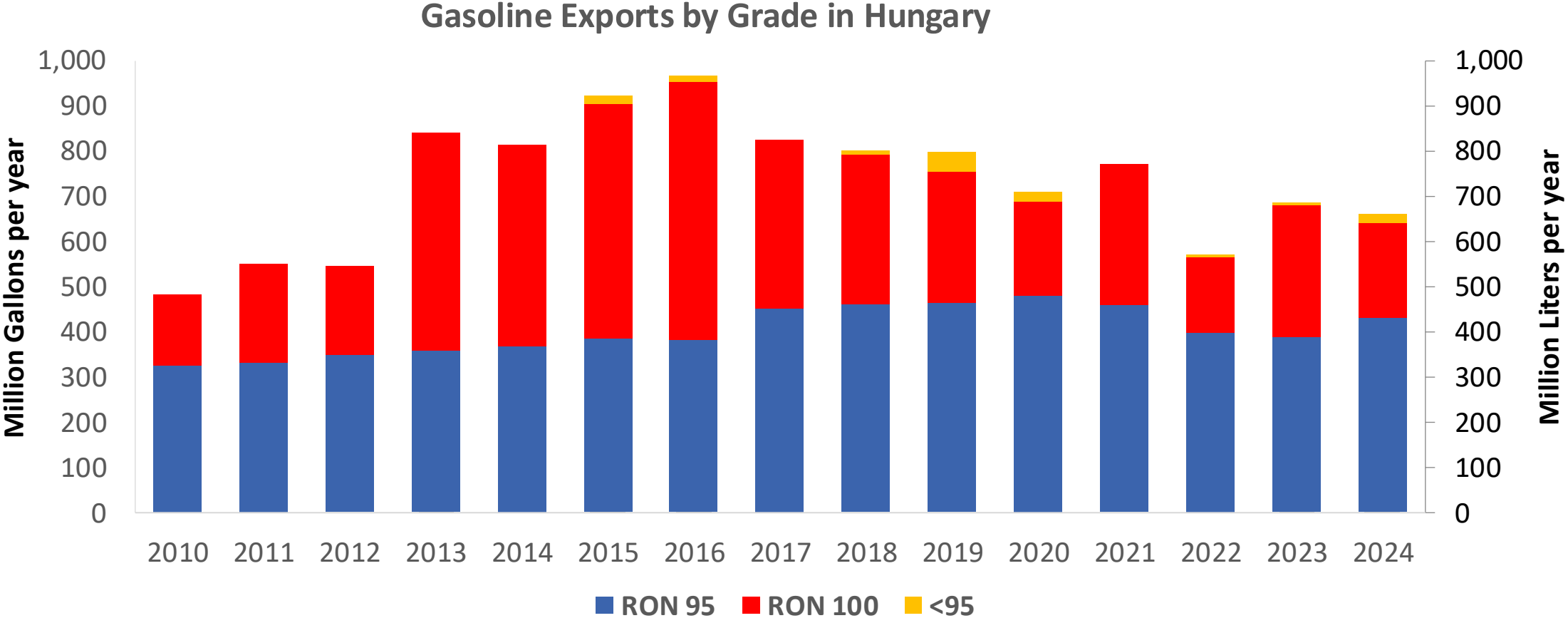
Source: Eurostat

Gasoline Imports in Hungary



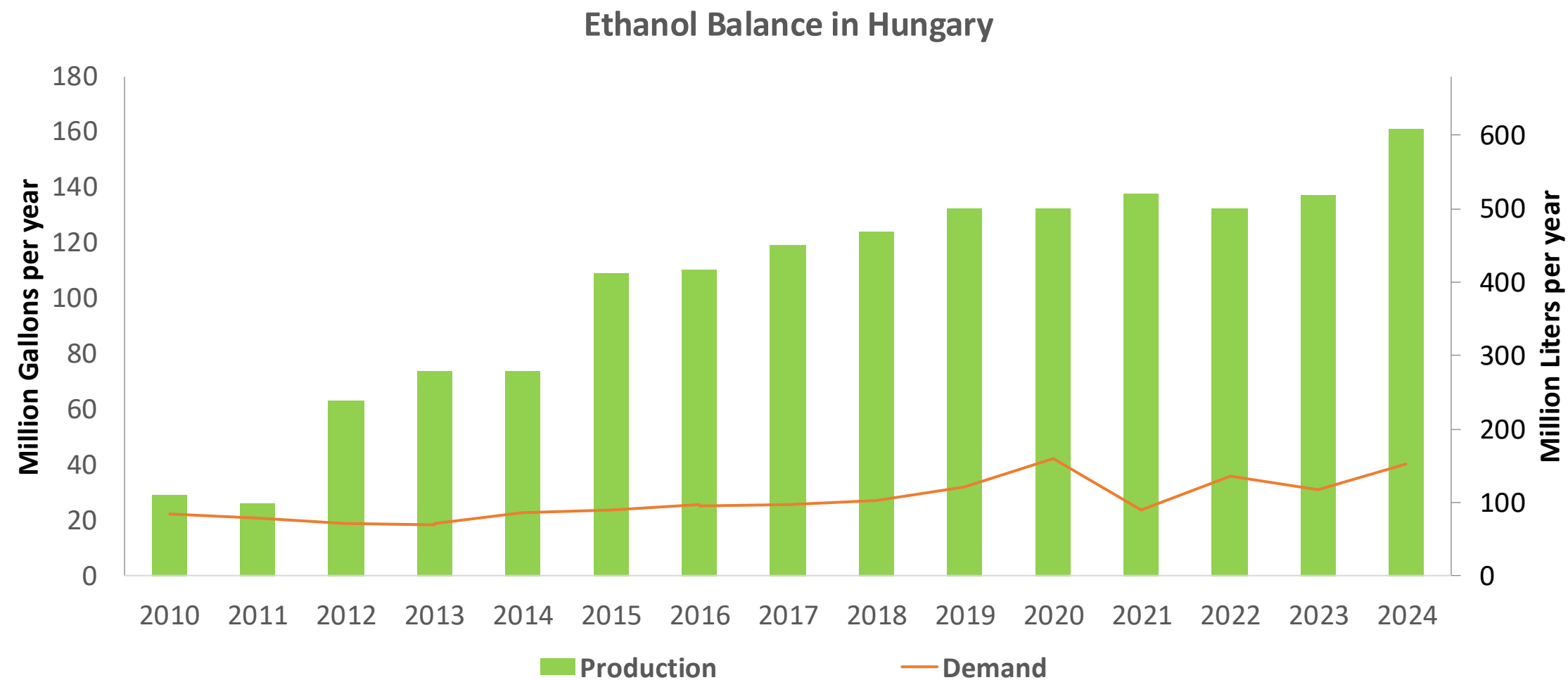
Source: Eurostat

Gasoline Exports in Hungary



Source: Eurostat

Ethanol Balance in Hungary



Source: Eurostat

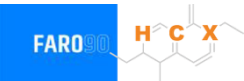
Gasoline Quality in Hungary

Name	EN 228:2025 (Euro 6); Country Level Fuel Ordinances							
Implementation Date	2025							
Applicability	All EU Countries + UK							
Grade	RON 95 E5		RON 95 E10		RON 98 E5		RON 98 E10	
	Min	Max	Min	Max	Min	Max	Min	Max
RON	95.0		95.0		98.0		98.0	
MON	> 85	> 88	> 85	> 88	> 85	> 88	> 85	> 88
AKI								
Benzene (%vol)	< 1							
Aromatics (% vol)	< 35							
Olefins (%vol)	< 18							
Lead (g/L)	< 0.005							
Manganese (mg/L)	< 2,0							
Sulfur (ppm)	10.0							
Oxygen (%w)*	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7
Ethanol (%vol)	< 5	< 10	< 5	< 10	< 5	< 10	< 5	< 10
PVR 37.8°C (psi)	<> 45 - 100 kPa in 6 Volatility Classes							
Overall Biofuels (%cal)	-							
Advanced Biofuels (%cal)	0.5							
Bioethanol in Gasoline Sales (%cal)	6.1							
GHG Emission Reduction (%)	-							

* Methanol 3%, Iso-propyl alcohol 12%, Ter-butyl alcohol 15%, Iso-butyl alcohol 15%, Ethers +5 Carbons 22%, Other Oxygenates 15%

Source: European Commission

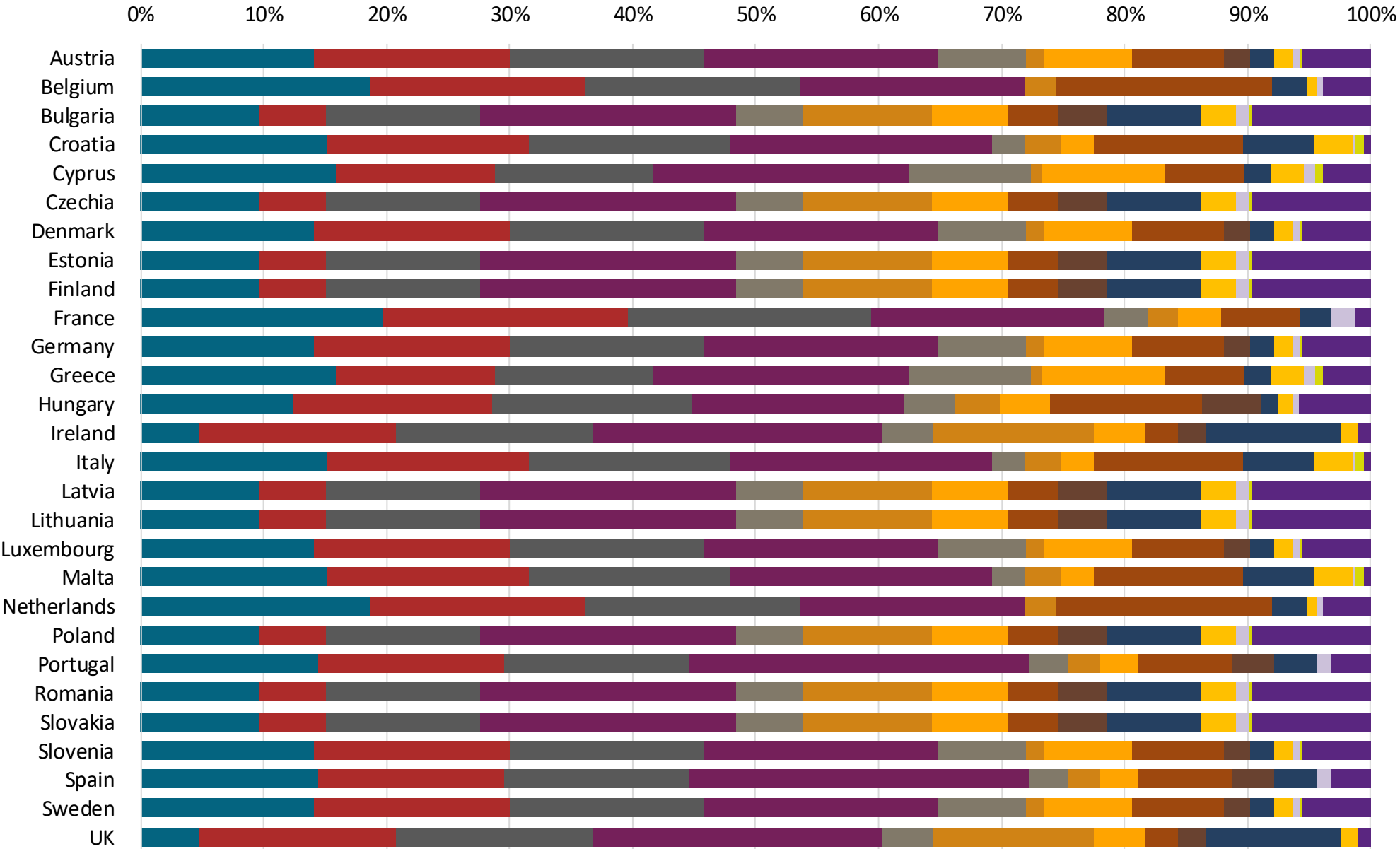
Gasoline Component Blending in Europe



Gasoline is a blend of a base gasoline and other components. Each component provides different properties to the final blend, for example, isomerates, alkylates and butanes increase the octane. The components commonly used in Europe are:

- Catalytic Gasoline
- Light Primary Naphta
- Heavy Primary Naphta
- Reformate
- Isopentane
- Alkylate
- Isomerate
- Hydrocracked Gasoline
- Coker Naphta
- n-Butane
- MTBE
- ETBE
- TAME
- Aromatics

Source: Faro90



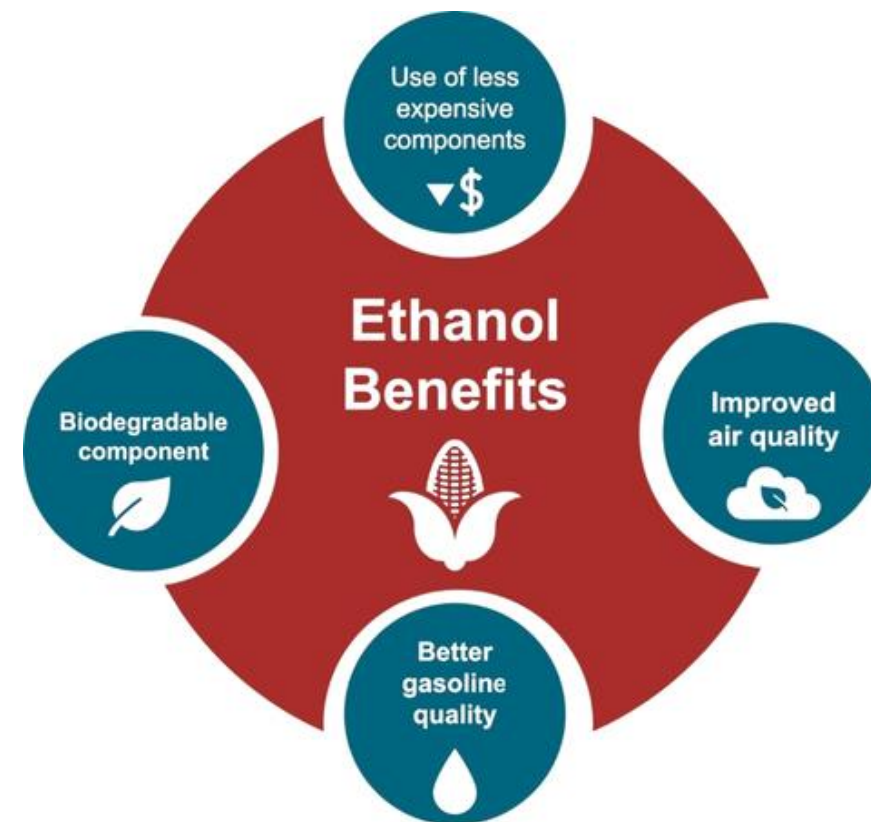
Gasoline Blending Optimization

In some parts of the world, ethanol is added to gasoline as a blending component. The advantages of ethanol include that it is a renewable fuel made of biomass; that it is an octane booster that helps to dilute sulfur; and that it allows the fulfillment of environmental objectives. To determine the optimal components to be blended with ethanol, a **blending model** was used. This model selects the components to add in the gasoline/ethanol blend based on:

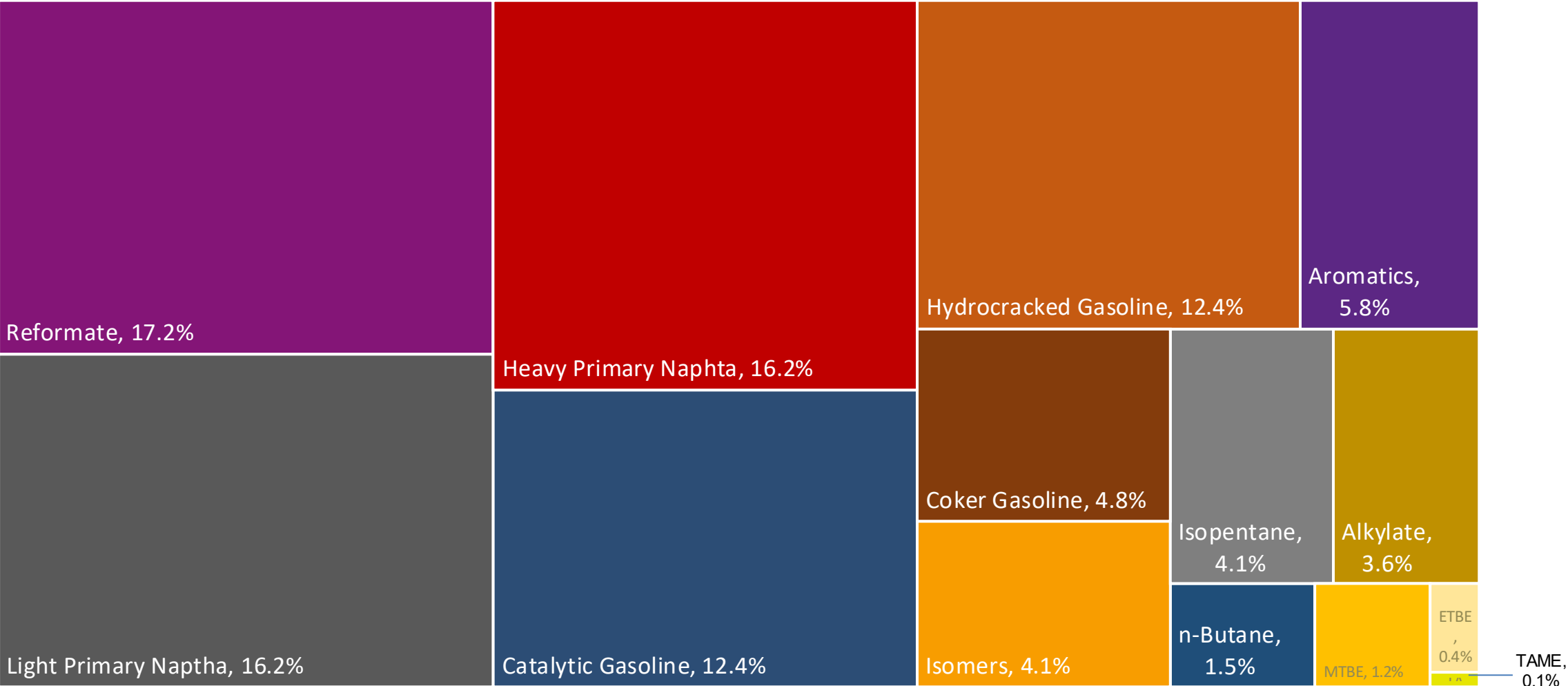
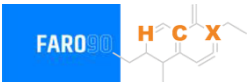
- Components prices,
- Properties each component affects,
- Quality parameters by country, and
- Component availability by country.

Through iterations, the model obtains the %v/v of the components to be blended with 10%, 15%, 20%, 25% and 30% of ethanol, in such a way that the final blend complies with the required properties of a finished gasoline by country.

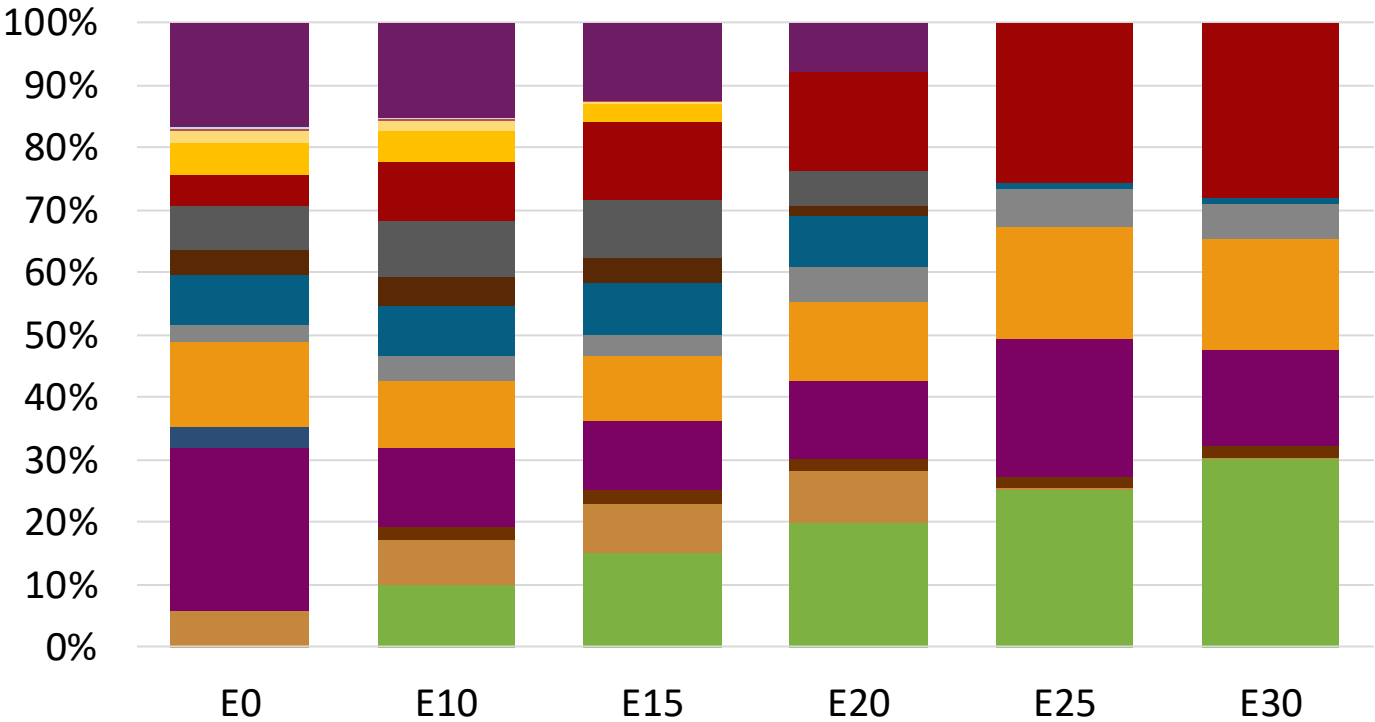
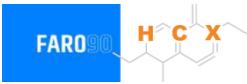
The blending model uses gasoline component spot average prices 2024 and provides fuel prices that do not include country distribution costs, local taxes and subsidies and import or gas station margins.



Hungary Available Blending Components



Hungary – Gasoline 95 – Constant Octane

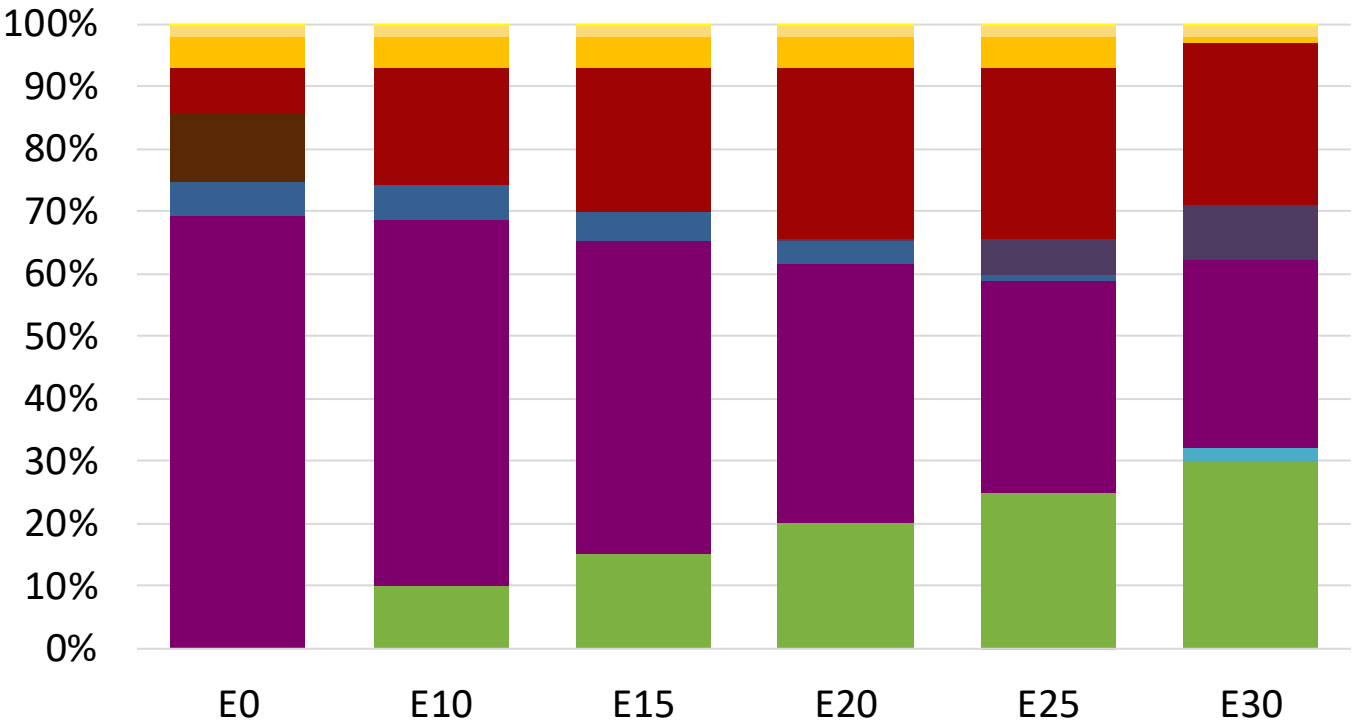
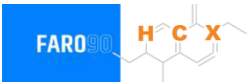


Average CIF 2024 Prices

Octane (RON)	95.0		95.0		95.0		95.0		95.0		96.0	
Price (US\$/gal)	\$	2.297	\$	2.168	\$	2.099	\$	2.021	\$	1.878	\$	1.841

Source: Faro90

Hungary - Gasoline 98 - Constant Octane

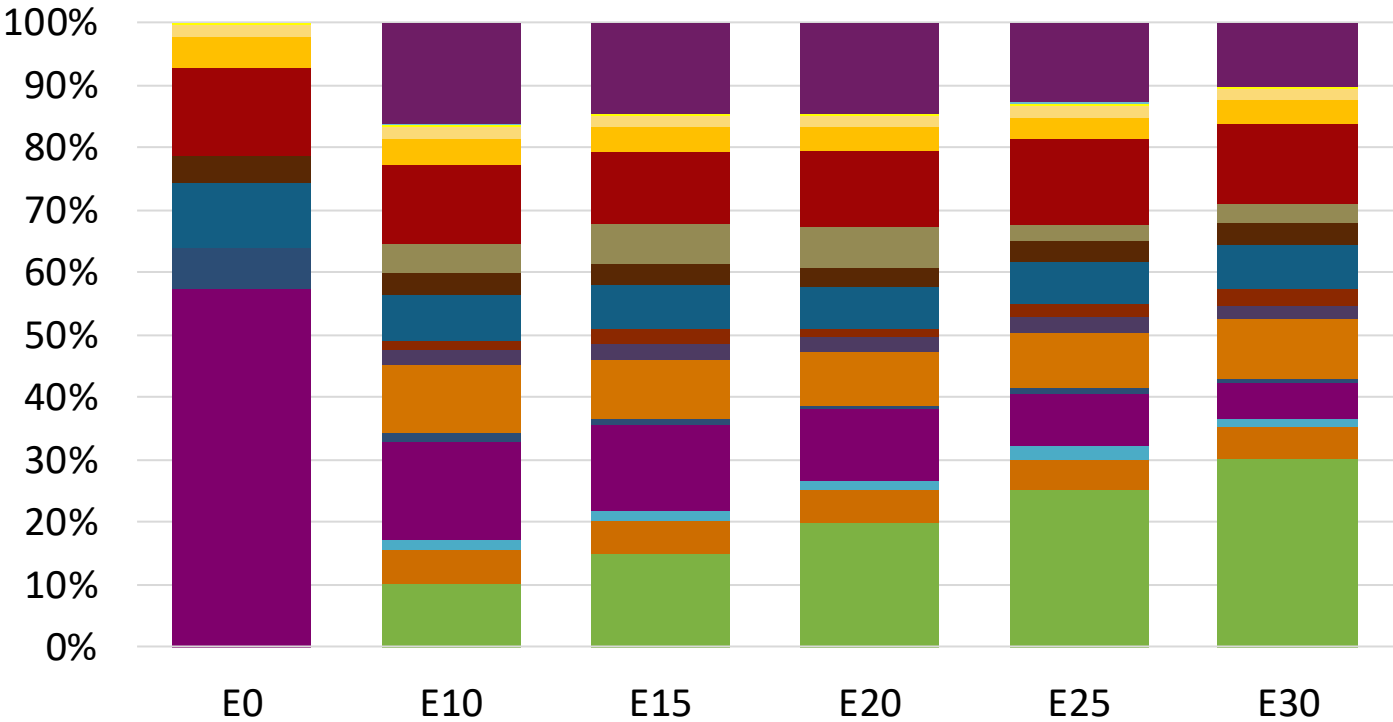
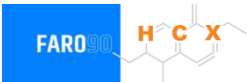


Average CIF 2024
Prices

Octane (RON)	98.0	98.0	98.0	98.0	98.0	98.0
Price (US\$/gal)	\$ 2.329	\$ 2.208	\$ 2.139	\$ 2.064	\$ 1.987	\$ 1.923

Source: Faro90

Hungary – Gasoline 95 – Increased Octane

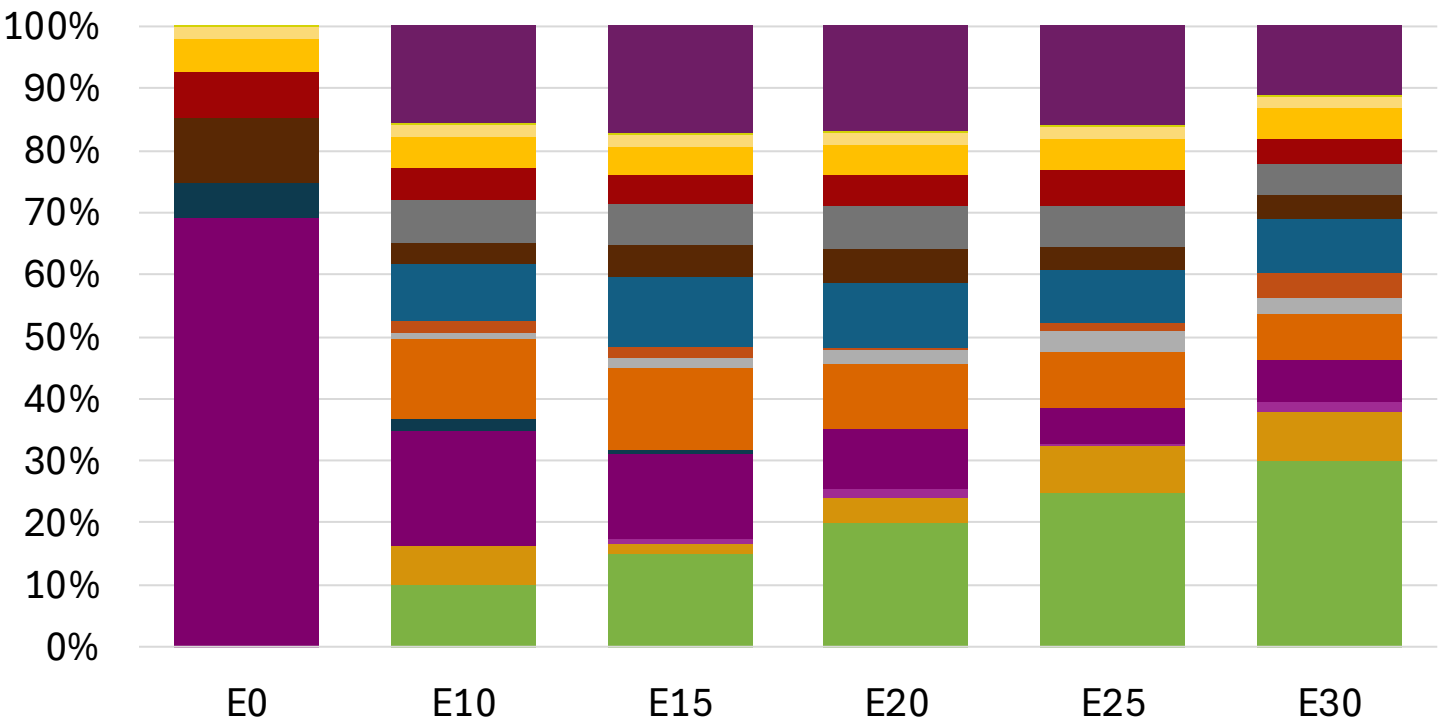


Average CIF 2024 Prices

Octane (RON)	95.0	95.7	97.2	98.8	100.3	101.8
Price (US\$/gal)	\$ 2.198	\$ 2.198	\$ 2.198	\$ 2.198	\$ 2.198	\$ 2.198

Source: Faro90

Hungary – Gasoline 98 – Increased Octane



Average CIF 2024 Prices

Octane (RON)		98.0	98.5	99.9	101.2	102.5	104.0
Price (US\$/gal)	\$	2.329	\$ 2.329	\$ 2.329	\$ 2.329	\$ 2.329	\$ 2.329

Vehicle Emission Impact for Ethanol Gasoline Blending

The model used in this analysis takes as a reference the [International Vehicle Emissions Model \(IVE\)](#).

The model uses the Base Emission Rates from IVE model, as well as its Adjustment Factors based on:

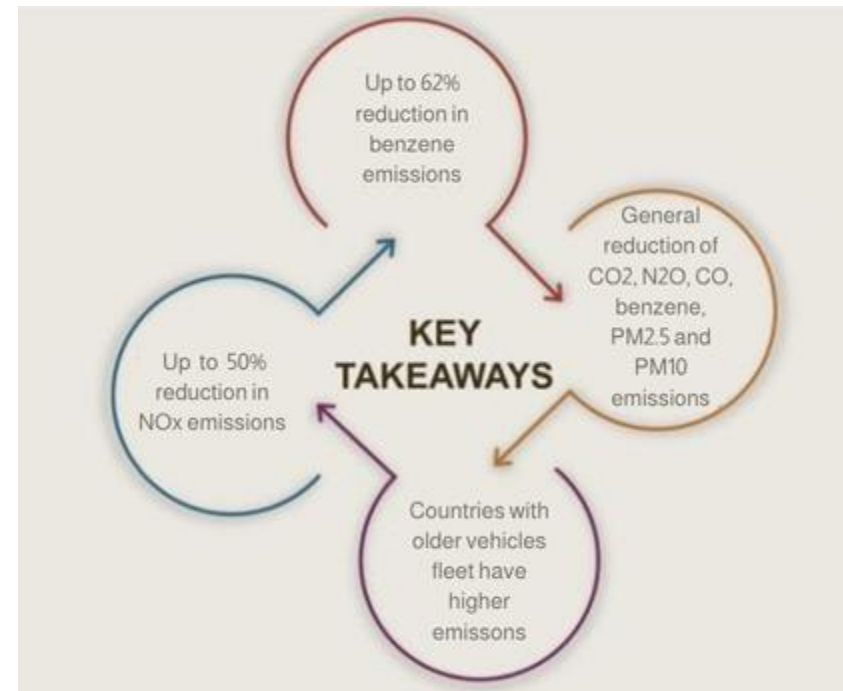
- Vehicle technology (cars, trucks, buses, motorcycles),
- Vehicle fleet average age,
- Average traveled distance per vehicle by country, as well as
- Geographical and climatic conditions (altitude, humidity, temperature).

Emissions of criteria pollutants, toxic pollutants, and greenhouse gases (GHG) were calculated and calibrated with emission inventories, using real gasoline quality data. The reduction rates for gasoline/ethanol blends were obtained from various sources (IPCC, US Grains, among others).

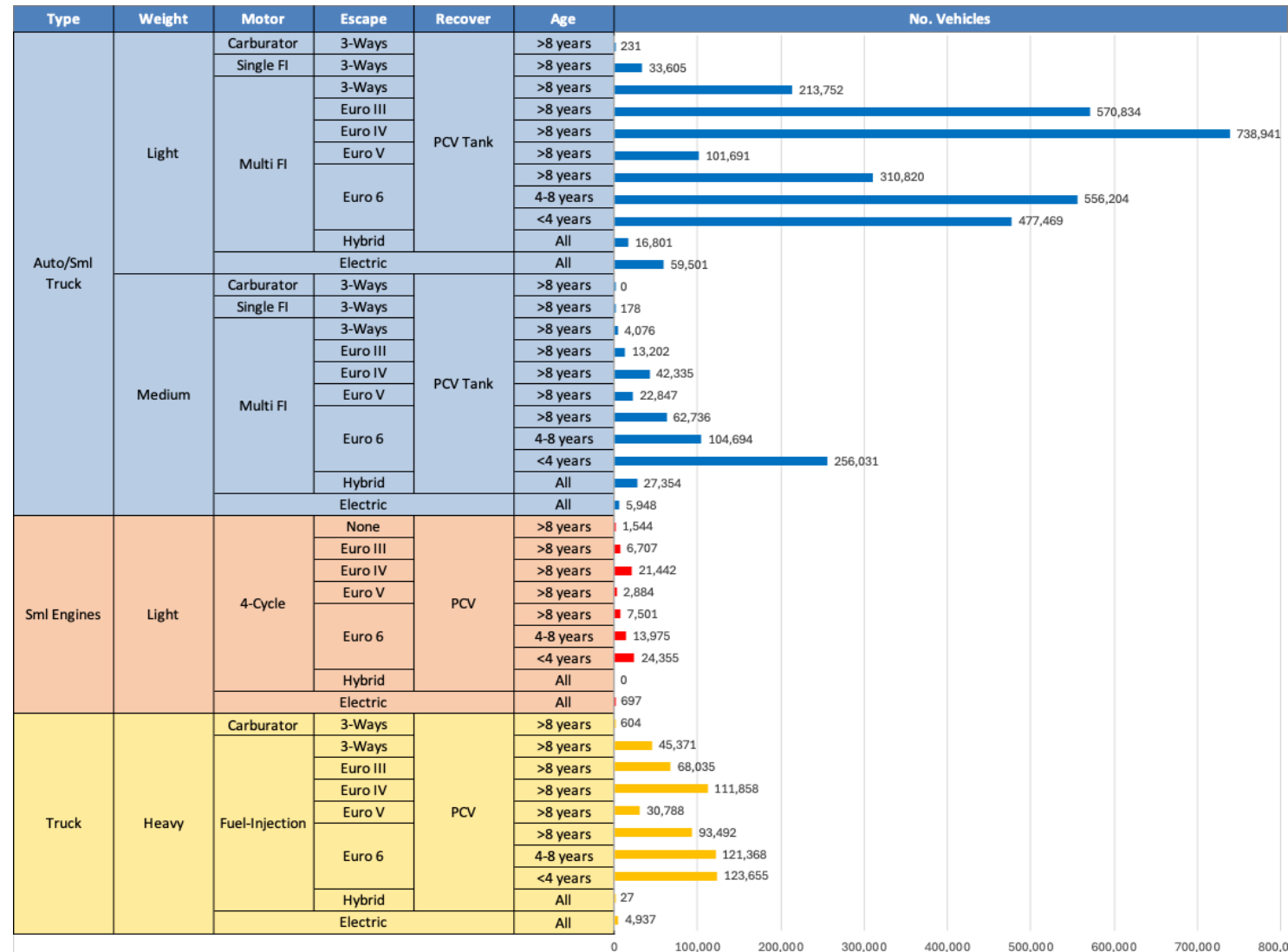
Emission estimations for different pollutants for gasoline and gasoline/ethanol blends (10%, 15%, 20%, 25% and 30% ethanol) were determined using the IVE Model. A comparison between the results and the European (Euro 6) requirements is made. Results are also compared with real emissions of the United States vehicle fleet*.

*Source: Bureau of transportation statistics.

Main Results



Gasoline Vehicle Fleet - Hungary

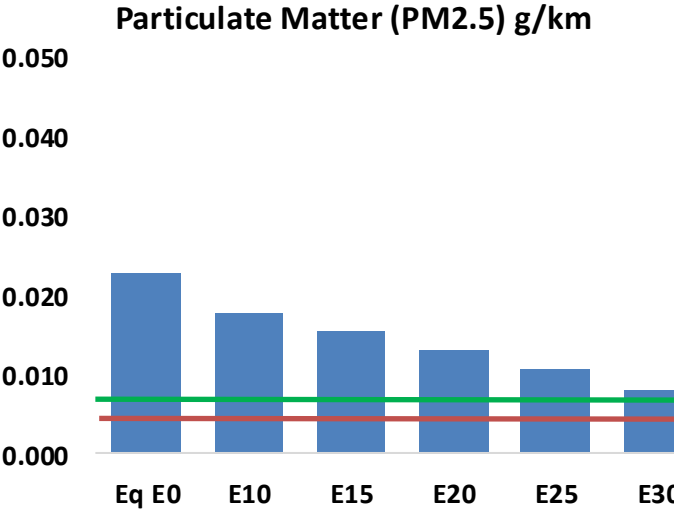
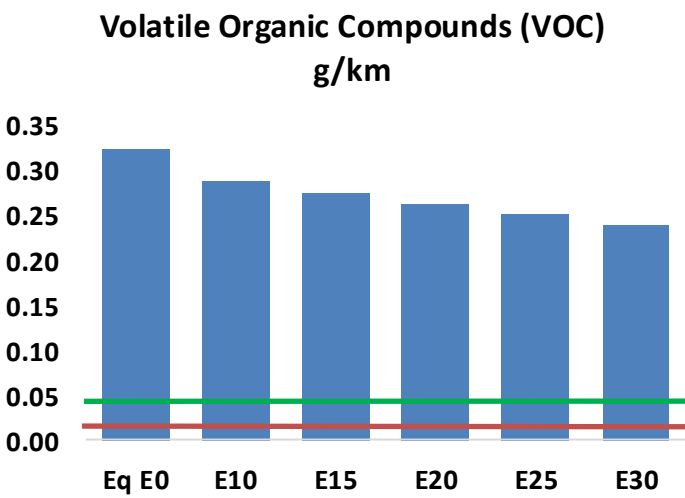
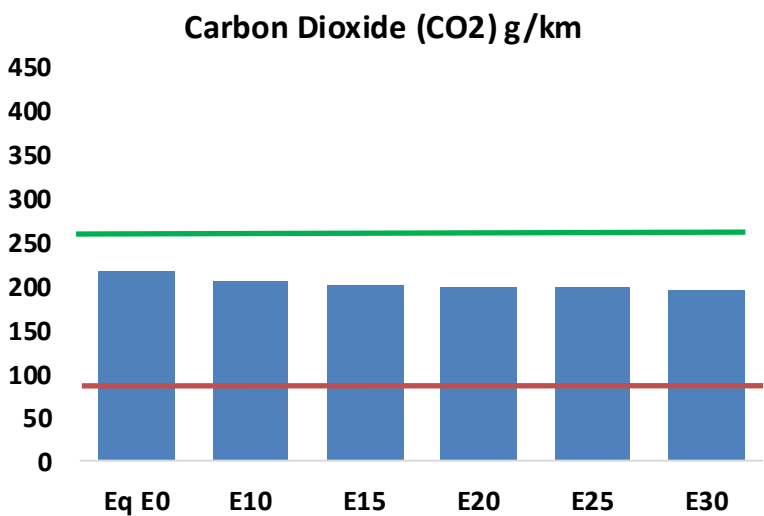
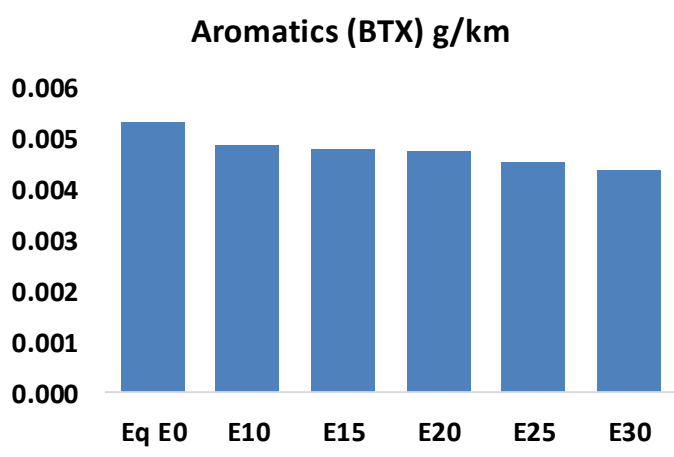
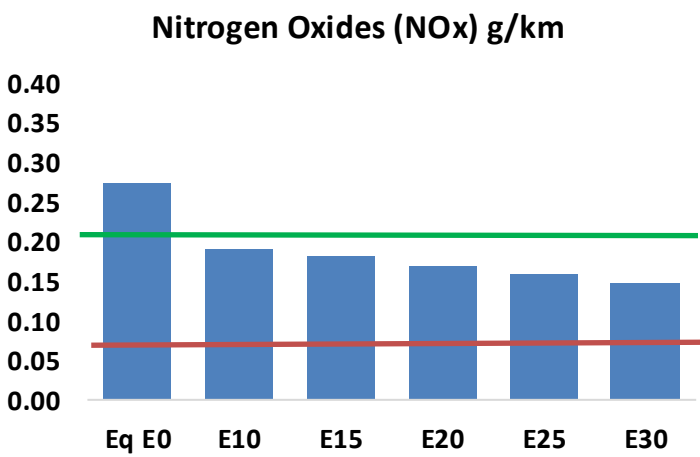
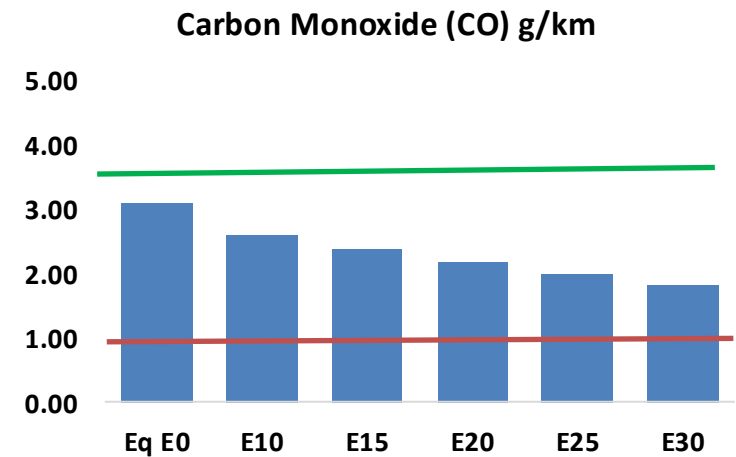
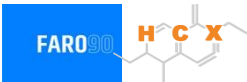


Vehicle Fleet: 4,298,491
Gasoline Vehicle Fleet: 2,317,855

Source: Eurostat, ACEA

Hungary – Vehicular Emissions

TIER III BIN 125 United States
Euro 6 Standard



Hungary – Vehicular Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	174,918	160,652	146,385	134,534	122,535	113,286	103,266	-16%	-30%
CO2	12,154,422	11,841,879	11,546,701	11,314,551	11,199,675	11,138,457	10,933,143	-5%	-8%
VOC	18,282	17,261	16,241	15,479	14,748	14,196	13,422	-11%	-19%
NOx	15,402	11,552	10,782	10,162	9,583	8,946	8,271	-30%	-38%
SOx	138	127	117	108	100	90	81	-15%	-28%
NH3	4,122	4,081	4,041	4,060	4,105	4,138	4,164	-2%	0%
N2O	188	187	186	187	191	193	195	-1%	2%
CH4	3,983	3,983	3,983	4,043	4,131	4,206	4,278	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	159	153	148	145	142	141	138	-7%	-10%
Acetaldehyde	342	393	576	877	1,195	1,366	1,614	68%	249%
Formaldehyde	1,074	1,044	1,217	1,429	1,497	1,657	1,803	13%	39%
Benzene	302	290	275	270	268	256	248	-9%	-11%
PM2.5	750	666	582	506	429	348	264	-22%	-43%
PM10	537	477	417	362	307	249	189	-22%	-43%

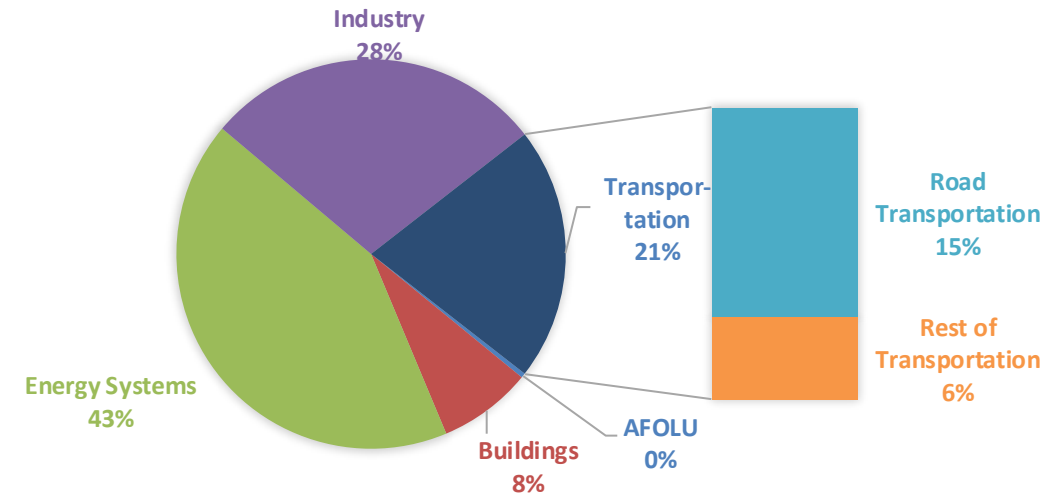
Life Cycle GHG Emissions

The Global Agenda for Climate Action calls for countries, cities and regions, businesses, and members of civil society around the world to achieve a low-carbon and climate-resilient economies in support of the Paris Agreement. Road Transportation accounts for 15% of total GHG emissions (IPCC,2023).

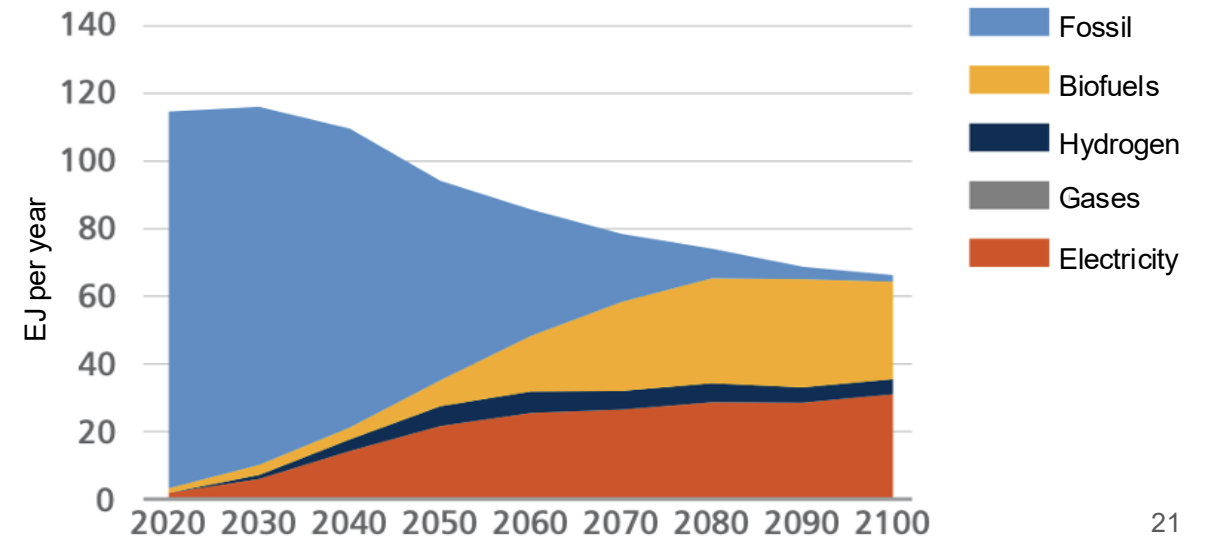
Ethanol gasoline blends are an internationally adopted practice that offer significant short-term benefits. Biofuels are an important solution to achieve reduction targets in the transportation sector.

Country GHG emission reductions are calculated using **RED II/III** and **GREET** life cycle emission methodologies and the **International Vehicular Emissions Model (IVE)**. Results are compared to the current country targets for 2035. Current 2024 country emission are those published by **IPCC**.

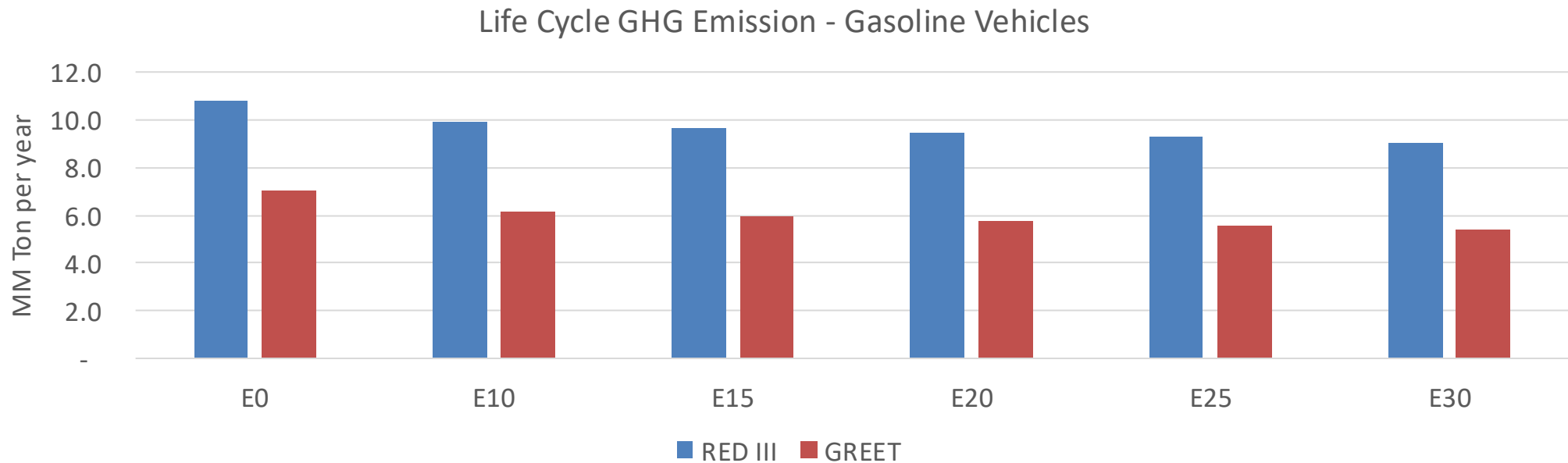
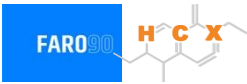
Current GHG Global Emissions Distribution



IPCC Transport Sector Sustainable Development Goals and climate policies Scenario, 2023



Hungary – Gasoline Vehicle Fleet Life Cycle GHG Emissions



Country Emissions 2024 (MMT/año)	64.6
Target @ 2035 (MMT/year)	36.4
Delta	28.2

%Reduction vs E0	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	12.2%	15.3%	17.8%	20.1%	23.3%
RED II/III	0.0%	8.4%	10.7%	12.5%	14.2%	16.4%

% Target@2035	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	3.0%	3.8%	4.4%	5.0%	5.8%
RED II/III	0.0%	3.2%	4.1%	4.8%	5.4%	6.3%

Source: Greet, RED II/III, IPCC, climateactiontracker.org, F90